

Experiment – 01

Home Task

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Question 1 (D):

Amazon wants to identify returning customers who made another purchase shortly after their first purchase. Write a PostgreSQL query to find all users who made another purchase within 1 to 7 days of an earlier purchase.

Requirements:

Compare purchases made by the same user only.

Ignore comparisons where the same transaction is matched with itself.

The second purchase must occur strictly after the first purchase (same-day purchases do not count).

The second purchase must occur within 7 days of the first purchase.

Display only the unique user_id values that satisfy the above conditions, sorted in ascending order.

Solution:

```
SELECT DISTINCT p1.user_id
FROM purchases p1
JOIN purchases p2
    ON p1.user_id = p2.user_id
    AND p2.purchase_date > p1.purchase_date
    AND p2.purchase_date <= p1.purchase_date + INTERVAL '7 days'
ORDER BY p1.user_id;
```

Explanation:

Self-joining the purchases table on user_id pairs up every purchase with every later purchase by the same user. The condition p2.purchase_date > p1.purchase_date excludes same-day and self-matches, while the upper bound of + INTERVAL '7 days' caps the gap at one week. DISTINCT then reduces the matching pairs down to unique user IDs.